

Independence of events

Def. A and B are ind.
 $\hookrightarrow$ independent

$$\text{iff } P(A \cap B) = P(A) \cdot P(B).$$

$\hookrightarrow$ if and only if

Notation : $A \perp B$ means A and B
are ind.

NOTE. Independence is a notion of
(no) relationship.

A_1, A_2 and A_3 are ind. iff

$$\begin{cases} P(A_1 \cap A_2) = P(A_1) \cdot P(A_2) \\ P(A_1 \cap A_3) = P(A_1) \cdot P(A_3) \\ P(A_2 \cap A_3) = P(A_2) \cdot P(A_3) \end{cases} \left(\begin{array}{l} P(A_1 \cap A_2 \cap A_3) \\ = P(A_1) \cdot P(A_2) \cdot P(A_3) \end{array} \right)$$

iv) $A_1, A_2, \dots, A_n$ are ind. iff any subsets of $A_1, A_2, \dots, A_n$ satisfy

$$P\left(\bigwedge_j A_j\right) = \prod_j P(A_j).$$

Ex. Roll 2 dices, one is in red, and the other is in green.

Let $A = \{ \text{the red dice} \geq 3 \}$

$B = \{ \text{the green dice} = 5 \}$

Is $A \perp B$?

$$\bullet P(A) = \frac{|A|}{|\Omega|} = \frac{4}{6}$$

$$\bullet P(B) = \frac{|B|}{|\Omega|} = \frac{1}{6}$$

$$\bullet P(A \cap B) = \frac{|A \cap B|}{|\Omega|} = \frac{4 \times 1}{6 \times 6}$$

$$\therefore P(A \cap B) = P(A) \cdot P(B)$$

$$\therefore A \perp B$$

Let $C = \{ \text{max of two dices} \geq 5 \}$

Is $A \perp C$? Red

$$\bullet P(A) = \frac{4}{6}$$

$$\bullet P(C) = \frac{20}{36}$$

$$\bullet P(A \cap C) = \frac{16}{36}$$

$$\therefore P(A \cap C) \neq P(A) \cdot P(C)$$

$$\therefore A \not\perp C$$

6						
5						
4						
3						
2						
1						
	1	2	3	4	5	6

Green

= A

= C

= A ∩ C

- NOTE :
- Disjoint events : $P(A \cup B) = P(A) + P(B)$
 - Ind. events : $P(A \cap B) = P(A) \cdot P(B)$

Some properties of independence :

- If A and B are disjoint, then A and B can NOT be ind. for $P(A) > 0$ and $P(B) > 0$.

Why ? $\because P(A \cap B) = P(\emptyset) = 0$

$$\therefore P(A \cap B) \neq P(A) \cdot P(B) > 0$$

- If $P(A) = 0$ or $P(A) = 1$, then A is ind. of any events.

Why ? (consider the case $P(A) = 0$, then

$$P(A \cap B) \leq P(A) = 0$$

$$\therefore P(A \cap B) = 0$$

$$\therefore P(A) \cdot P(B) = 0$$

$$\therefore P(A \cap B) = P(A) \cdot P(B) \iff A \perp B$$

When $P(A) = 1$, then we have

$$P(A \cap B) = P(B)$$

$$= P(A) \cdot P(B)$$

$$\therefore A \perp B.$$

• If $A \subseteq B$, then $A \not\perp B$
unless $P(A) = 0$ or $P(B) = 1$.

Why? $P(A \cap B) = P(A) \neq P(A) \cdot P(B)$

$$\therefore A \not\perp B.$$

$\nabla A \equiv B$ implies

$$\left\{ \begin{array}{l} A^c \equiv B \\ A \equiv B^c \\ A^c \equiv B^c \end{array} \right.$$